

Lab-3

ID	Name
2304106	Gasser Ibrahim Shalaby
2304071	Mohamed Osama

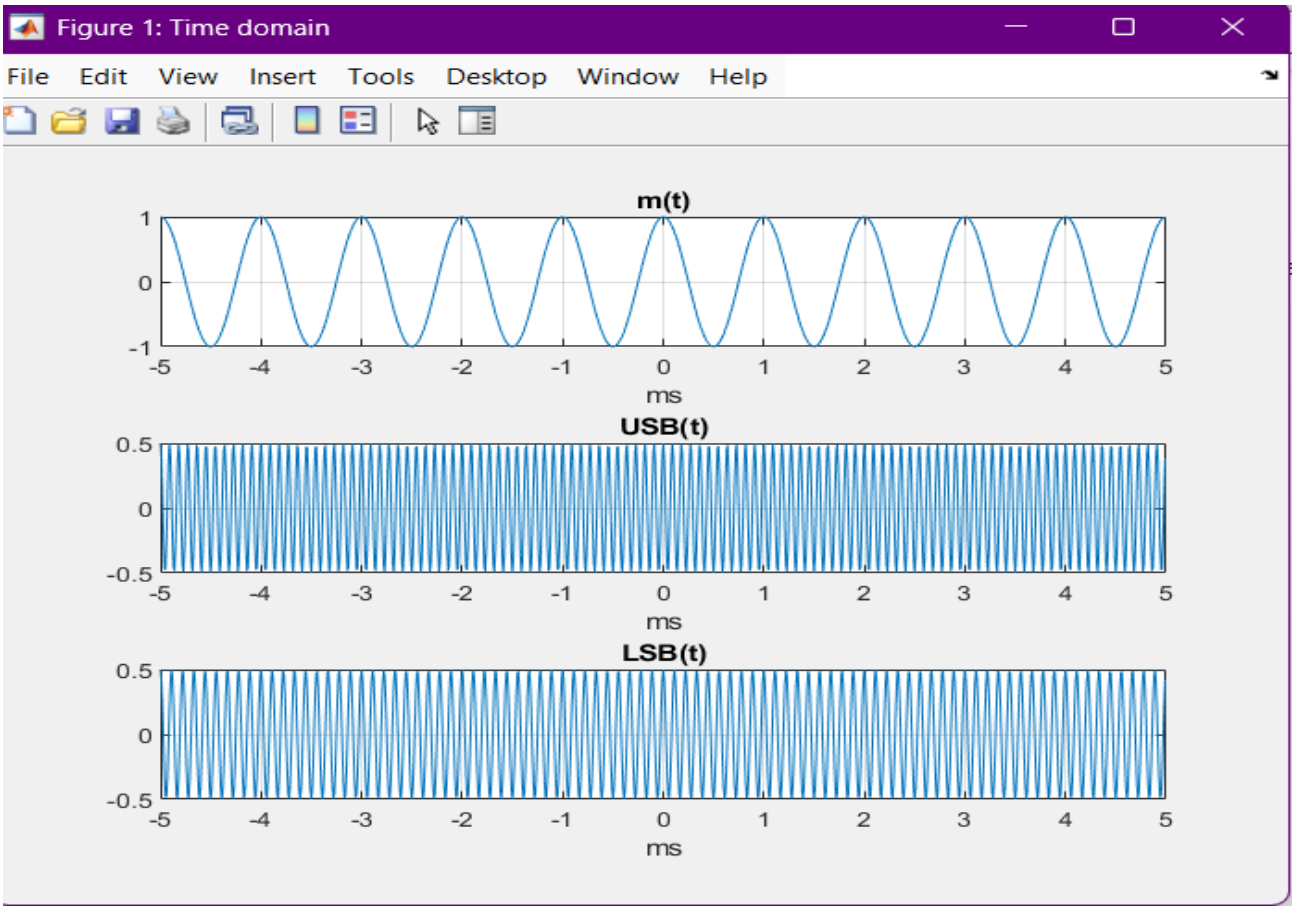
Part 1: Single Sideband Suppressed Carrier (SSB-SC)

***Q1*: DSB-SC.** $m(t)\cdot c(t) = \frac{1}{2}\cos(2\pi\cdot 9k\cdot t) + \frac{1}{2}\cos(2\pi\cdot 11k\cdot t)$. Its spectrum has lines at ±9 kHz and ±11 kHz.

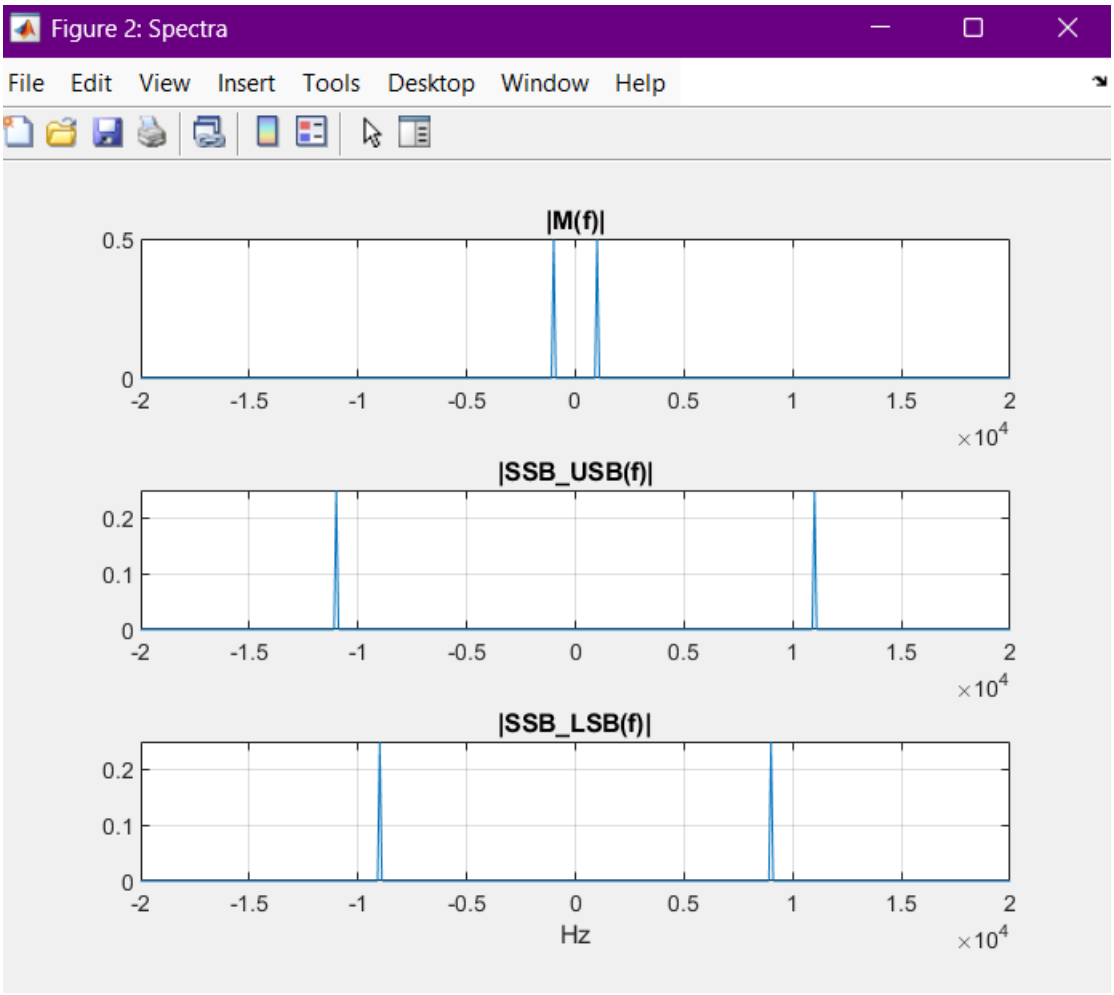
***Q2–3:* USB and LSB by manual frequency-domain filtering.** The DSB-SC spectrum is multiplied by an ideal mask and transformed back with the inverse FFT.

- USB: keep $|f| \geq Fc$ and set all bins with $|f| < Fc$ to zero. The result is $\frac{1}{2}\cos(2\pi\cdot 11k\cdot t)$.
- LSB: keep $|f| < Fc$ and set all bins with $|f| \geq Fc$ to zero. The result is $\frac{1}{2}\cos(2\pi\cdot 9k\cdot t)$.

***Q4*: time-domain plots.** m(t) is a 1 kHz cosine of amplitude 1 V. USB(t) and LSB(t) are constant-amplitude cosines (½ V) at 11 kHz and 9 kHz respectively, because a single-tone message has only one component in each sideband.



***Q5*: magnitude spectra over [−20000, 20000] Hz.** |M(f)| has lines at ±1 kHz only. |SSB_USB(f)| has lines at ±11 kHz only and |SSB_LSB(f)| has lines at ±9 kHz only, so each sideband is cleanly removed.



***Q6:* transmission bandwidth.** The SSB signal occupies only one sideband, so B_SSB = W = Fm = 1 kHz (a single line at 11 kHz for the USB). DSB-SC has lines at 9 and 11 kHz, so B_DSB = 2W = 2 kHz. **SSB needs half the bandwidth of DSB-SC.**

***Q7*: SSB using the Hilbert transform.** For m(t) = cos(2πFm t), the Hilbert transform is m̂(t) = sin(2πFm t). Then

$$USB(t) = m(t)\cos(2\pi Fc\ t) - \hat{m}(t)\sin(2\pi Fc\ t) = \cos(2\pi(Fc+Fm)t)$$

$\hat{m}(t)=Hm(t)=H(\cos 2000\pi t)=\sin(2000\pi t)$

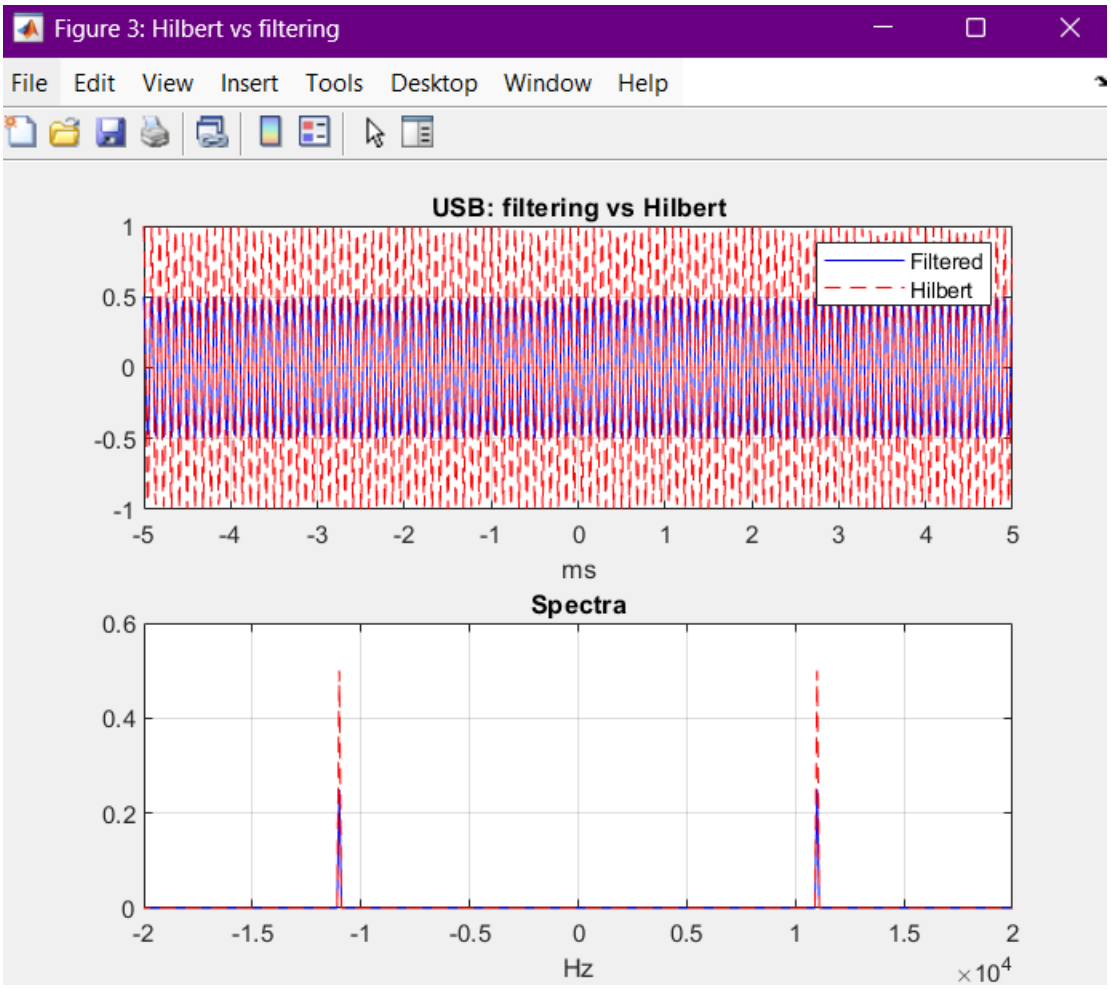
$USB(t)= (\cos 2000\pi t)\ .\ (\cos 2000\pi t)- \sin(2000\pi t\ ).\ \sin(2000\pi t)$

$USB(t)= (\cos 22000\pi t)$

$Fc+fm=10000+1000=11000Hz=11kHz$

This has amplitude 1. The filtered sideband has amplitude ½, because the DSB-SC product carries a factor ½. The standard form includes the factor: USB(t) = ½[m cos(ωct) − m̂ sin(ωct)].

***Q8*: comparison with the frequency-filtered signal.** The two signals coincide. The maximum difference is about 1.2×10^{-11} and the spectra overlap exactly (Figure 3). Both give a single line at 11 kHz. The Hilbert method needs no sharp sideband filter, which is its practical advantage; filtering needs an ideal (unrealisable) filter.

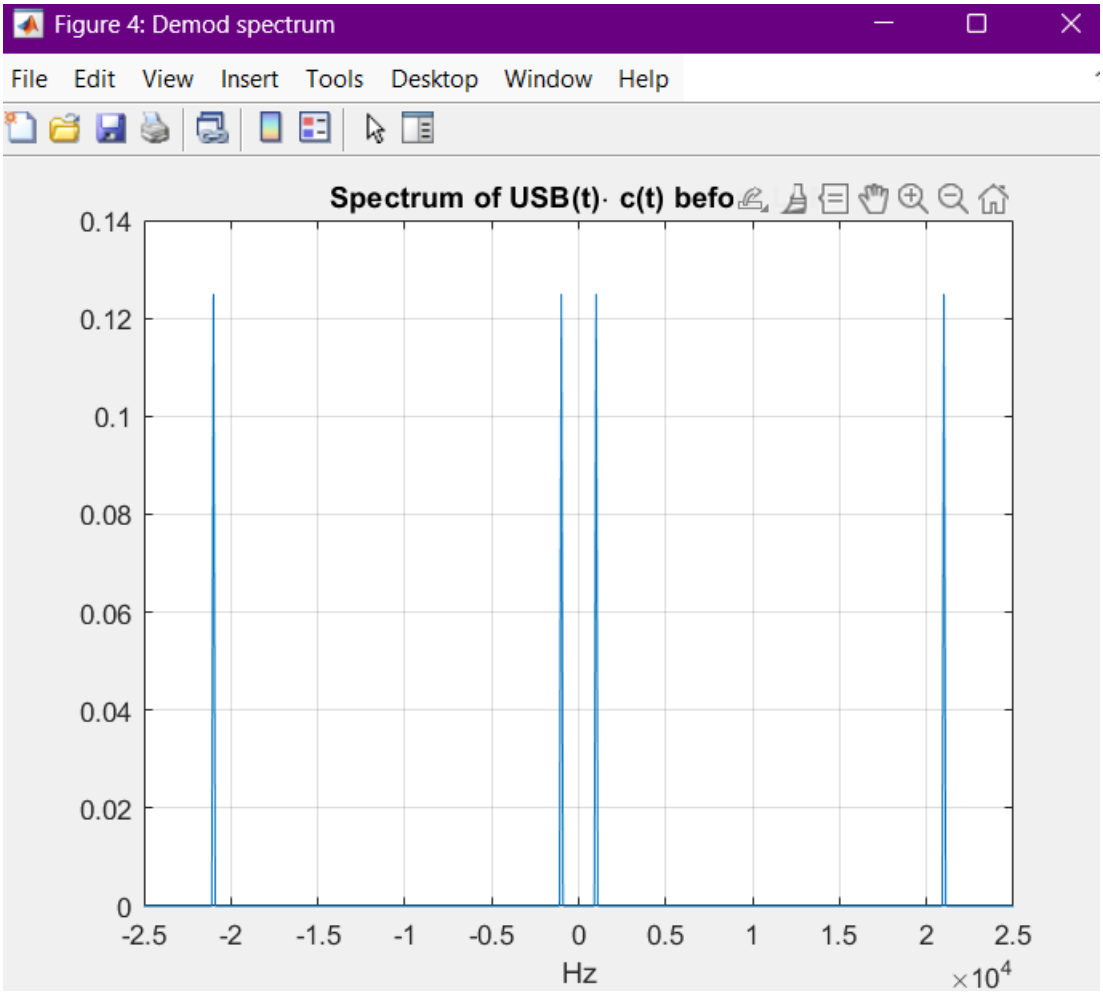


Coherent demodulation and filtering

***Q9*: coherent detection.** Multiplying by the synchronised carrier:

$$USB(t)\cdot c(t) = ¼\cos(2\pi\cdot1k\cdot t) + ¼\cos(2\pi\cdot21k\cdot t)$$

***Q10*: spectrum before filtering.** There are two pairs of lines: the message at ±1 kHz and the 2Fc image at ±21 kHz, each of single-sided amplitude ¼.



***Q11*: effect of a phase error θ and a frequency offset Δf.** Let the receiver oscillator be cos(2π(Fc+Δf)t + θ). For a USB signal with message m(t) and Hilbert transform m̂(t), the low-pass output is

$$\tfrac{1}{4}[\, m(t)\cos(\Delta\omega t+\theta) + \hat{m}(t)\sin(\Delta\omega t+\theta)\,] \;\longrightarrow\; \tfrac{1}{4}\cos(2\pi(Fm-\Delta f)t-\theta) \quad \textit{for a tone message}$$

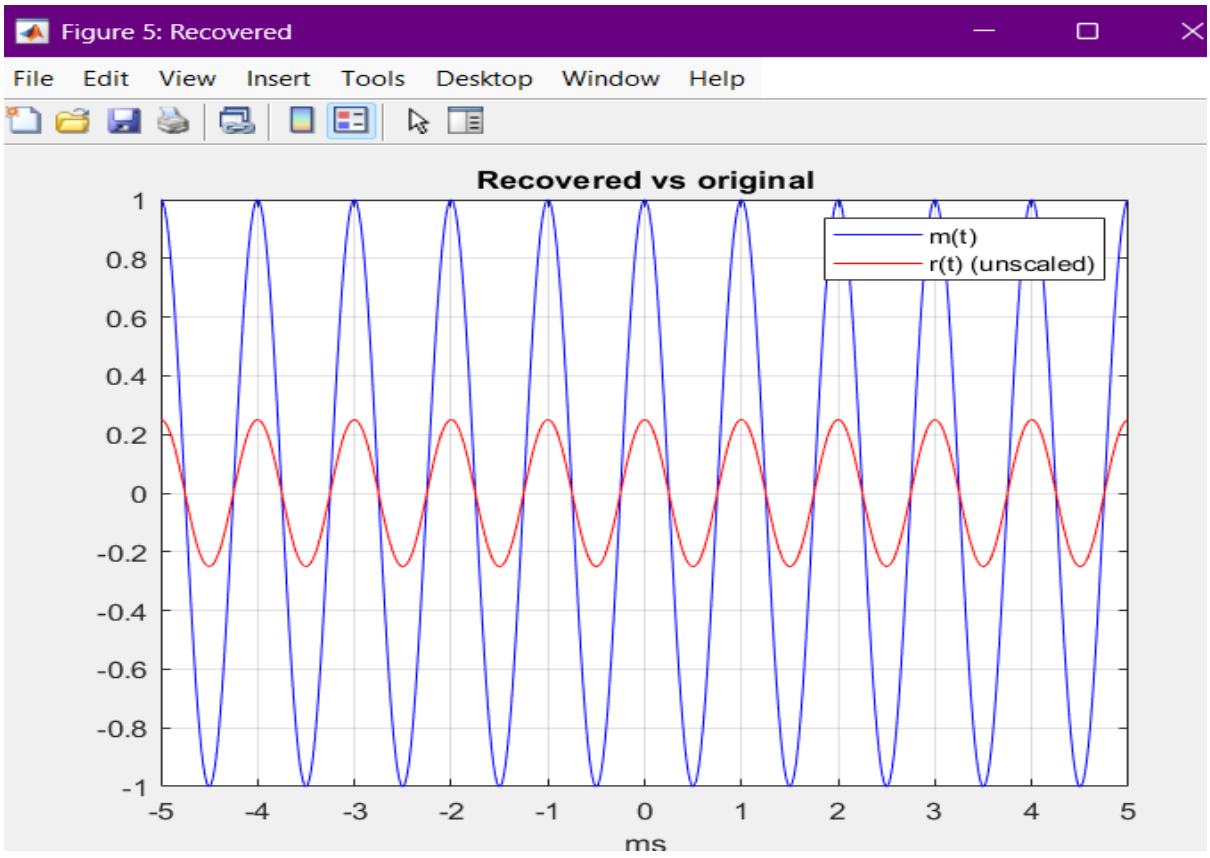
- **Phase error (Δf = 0):** the output is a phase-shifted copy of the message (m cosθ + m̂ sinθ). In SSB the amplitude is not reduced, the effect is phase distortion. Simulated: the amplitude stays 0.25 for θ = 0°, 45° and 90°. In DSB-SC the output is multiplied by cosθ, so it fades and vanishes at θ = 90° (simulated amplitude ≈ 0).
- **Frequency offset:** every message component is shifted by Δf (down for USB, up for LSB), so the pitch changes and the output is distorted (the "Donald Duck" effect on speech). Simulated: for Δf = 200 Hz the recovered tone appears at 800 Hz instead of 1000 Hz.
- **Practical impact:** the local oscillator must be locked in frequency and phase to the carrier. This needs a PLL or Costas loop, or a transmitted pilot tone. Even small offsets are audible, which makes SSB-SC receivers more complex than envelope receivers.

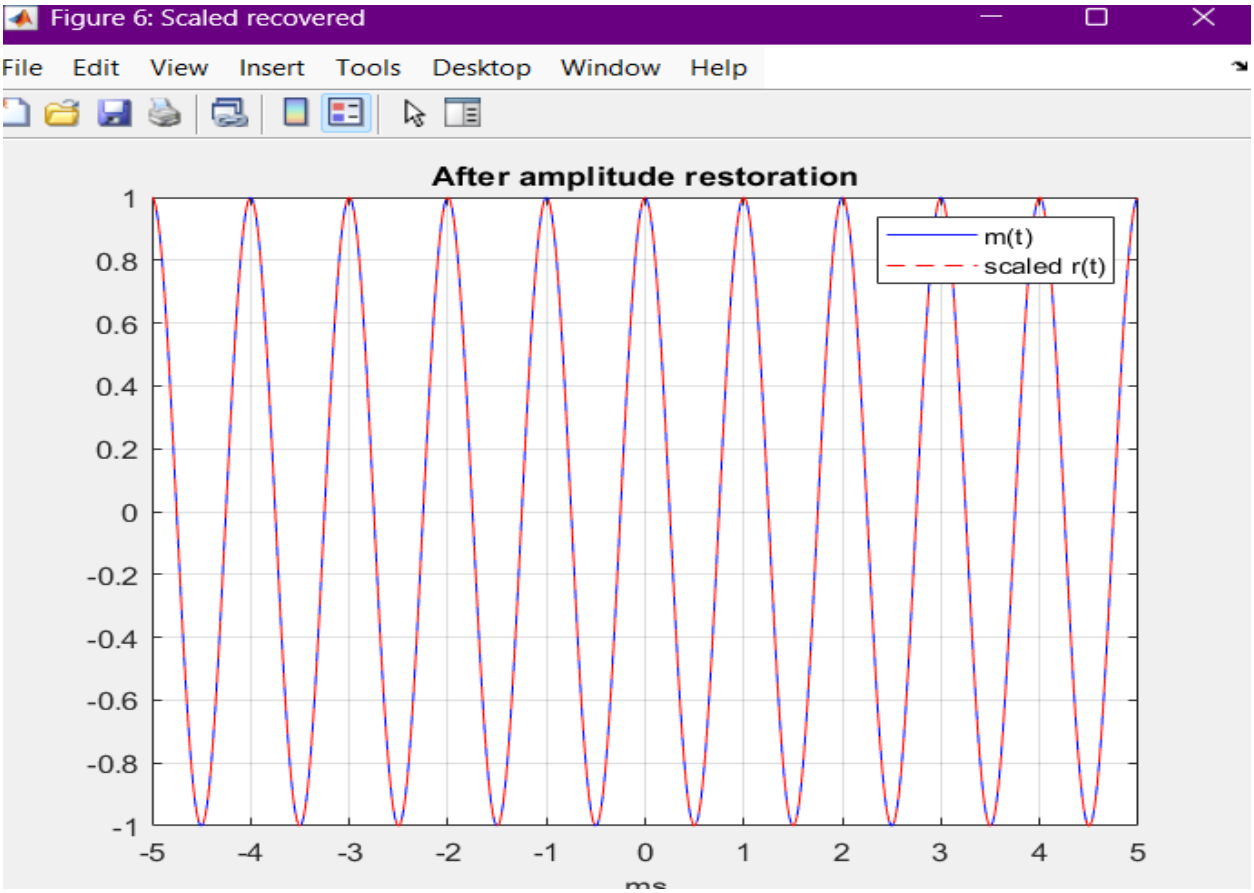
***Q12*: manual low-pass filter, Fc_cut = 1000 Hz.** The filter was built by hand in the frequency domain (an ideal mask, $|f| \leq 1$ kHz, applied to the FFT), without any built-in filter function. It removes the 21 kHz term and keeps the 1 kHz term. r(t) is a clean 1 kHz cosine in phase with m(t) but smaller in amplitude (Figure 5, top).

***Q13*: amplitude comparison and scaling factor.** The recovered amplitude is 0.25 V (measured 0.2500), compared with 1 V for m(t). The factor is:

$$r(t) = (Am\ Ac / 4)\ cos(2\pi Fm\ t) \quad \Rightarrow \quad scaling\ factor = 4/Ac = 4$$

One factor of ½ comes from the SSB signal (one sideband carries half the DSB-SC amplitude) and one from the product of two cosines. After multiplying by 4, r(t) overlays m(t) exactly (Figure 5, bottom). Note: a real filter has gain 0.5 at its cutoff frequency, so a practical filter whose cutoff equals Fm would need a factor of 8; with the cutoff placed slightly above Fm the factor is 4.





***Q14*: why envelope detection fails for SSB-SC but works for SSB-TC.**

The envelope of the SSB-SC signal is

$$\tfrac{1}{2}\sqrt{(\, m(t)^2 + \hat{m}(t)^2\,)}$$

which is not proportional to m(t) (for a single tone it is a constant, so the message is lost). In SSB-TC the large carrier A·c(t) is added in phase, so the envelope becomes

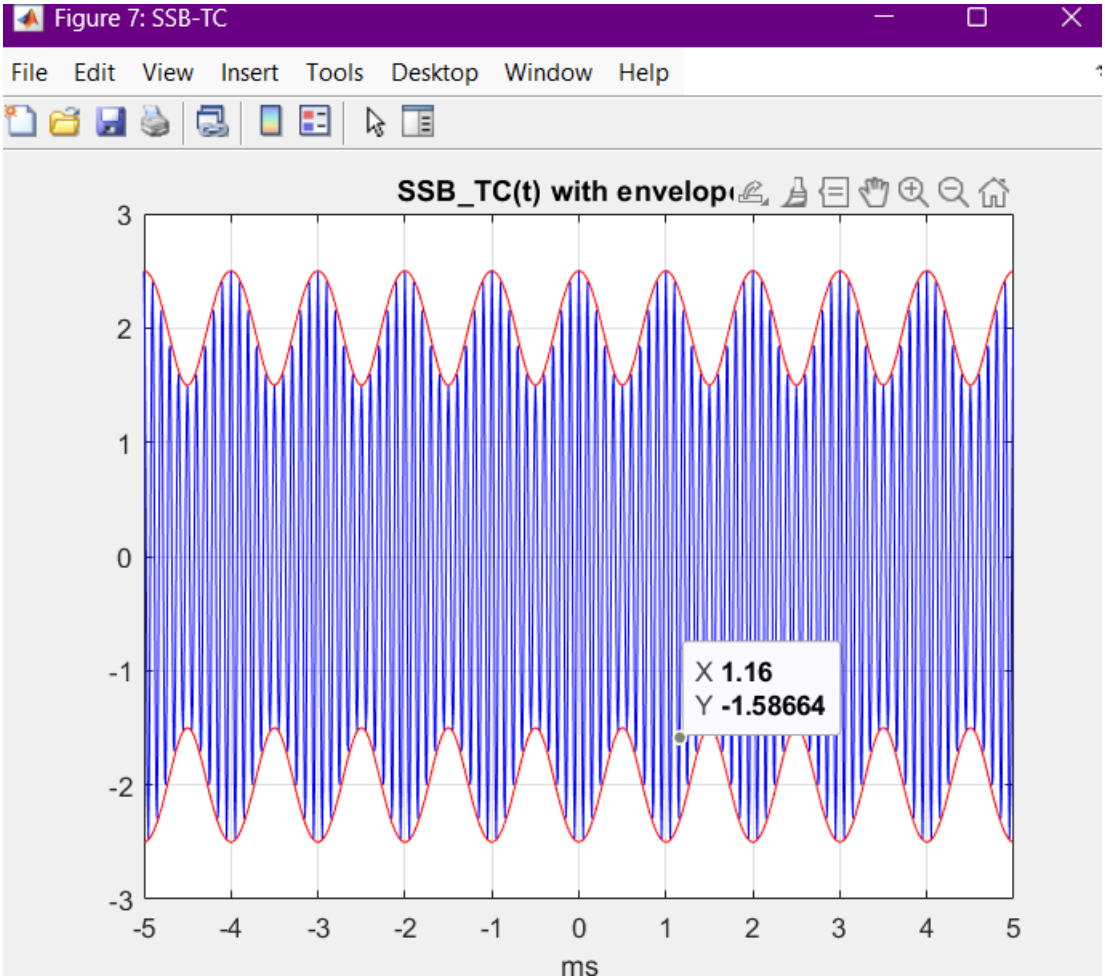
$$\sqrt{(\,(A + m(t)/2)^2 + (\hat{m}(t)/2)^2\,)} \;\approx\; A + m(t)/2 \quad \text{for } A \gg |m|$$

which contains the message on top of a DC level. The carrier acts as the phase reference that makes envelope detection valid.

Part 2: Single Sideband Transmitted Carrier (SSB-TC)

Q1(SSB-TC(t) = A·c(t) + USB(t) with A = 2 V, SSB-TC(t) = 2cos(2π·10k·t) + ½ cos(2π·11k·t).

Q2)



Q3) The envelope is the magnitude of the analytic signal, env(t) = |SSB_TC(t) + j·Hilbert{SSB_TC(t)}|.Envelope detection and comparison

Step 3: envelope extraction. The envelope is the magnitude of the analytic signal, env(t) = |SSB_TC(t) + j·Hilbert{SSB_TC(t)}|.

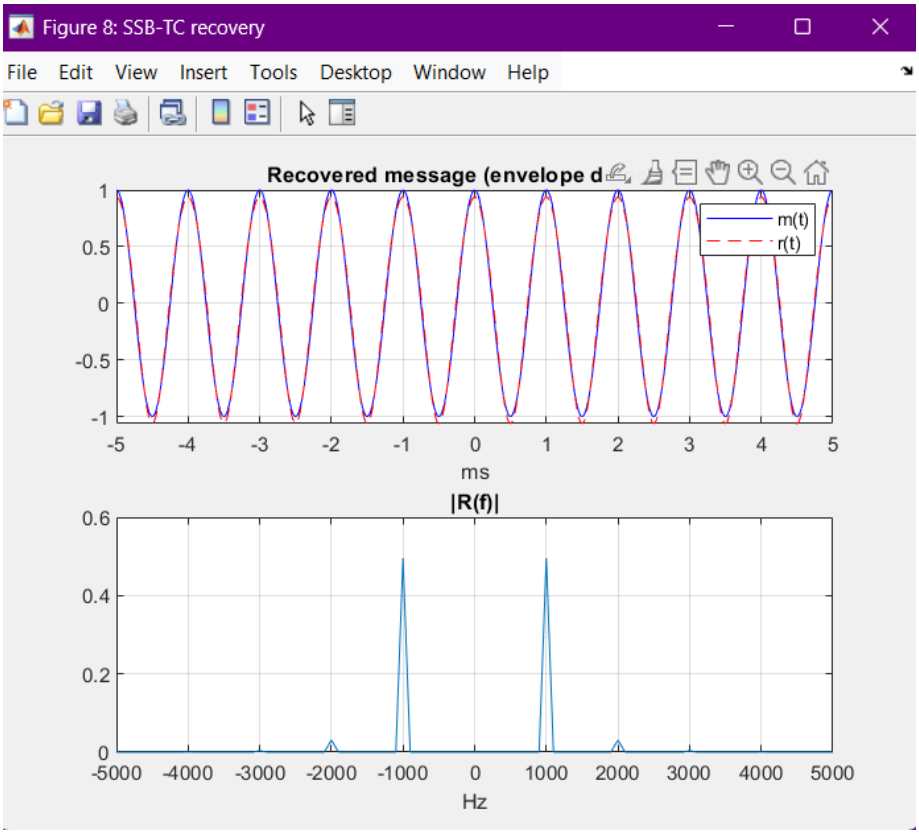
Step 4: removing the DC offset. The envelope is A + m/2 (plus small distortion), so r(t) = 2·(env(t) − mean(env)). The factor 2 restores the amplitude because the USB carries m(t)/2.

Step 5: minimum condition on A. To avoid envelope distortion the carrier must dominate the message, so the minimum condition is **A ≥ max|m(t)| = Am**, the same as the no-overmodulation condition for DSB-TC. Because the m̂ term makes the SSB envelope slightly nonlinear, the distortion keeps decreasing as A grows (A ≫ Am/2 for a faithful copy). Simulated maximum recovery error:

A (V)	0.25	0.5	1	2
Max |r(t) − m(t)|	0.56	0.27	0.13	0.063

With A = 2 V and Am = 1 V the condition is satisfied with margin (A/Am = 2).

Q6:



***Q7*: comparison of DSB-SC, DSB-TC and SSB.**

	Power efficiency	Bandwidth	Receiver complexity
DSB-SC	High: no power is spent on the carrier	2W (2 kHz here)	Complex: coherent detector with carrier recovery (PLL / Costas loop)
DSB-TC (AM)	Low: $\eta = \mu^2/(2+\mu^2)$, at most 33% at $\mu = 1$	2W (2 kHz here)	Very simple: envelope detector, no synchronisation
SSB-SC	Highest: only one sideband is transmitted, no carrier	W (1 kHz here)	Complex: coherent detection with tight synchronisation, plus a sharp filter or Hilbert phase-shift modulator at the transmitter
SSB-TC	Low: the carrier takes most of the power	W (1 kHz here)	Simple envelope detector, but more complex transmitter and some residual distortion

```
clear; clc; close all;

% ---------------- Parameters ----------------
Am = 1;  Fm = 1000;
Ac = 1;  Fc = 10000;
Fs = 100000;
t  = -0.005:1/Fs:0.005-1/Fs;
N  = length(t);
f  = (-N/2:N/2-1)*Fs/N;
tm = t*1e3;

m = Am*cos(2*pi*Fm*t);
c = Ac*cos(2*pi*Fc*t);
spec = @(x) fftshift(fft(x))/N;

%% ================= PART 1: SSB-SC =================
% 1) DSB-SC
dsb = m.*c;

% 2) USB by manual frequency-domain filtering (keep |f| >= Fc)
S      = fftshift(fft(dsb));
usb    = real(ifft(ifftshift(S.*(abs(f) >= Fc-1))));

% 3) LSB (keep |f| < Fc)
lsb    = real(ifft(ifftshift(S.*(abs(f) <  Fc-1))));

% 4) Time-domain plots
figure('Name','Time domain');
subplot(3,1,1); plot(tm,m);   grid on; title('m(t)');   xlabel('ms');
subplot(3,1,2); plot(tm,usb); grid on; title('USB(t)'); xlabel('ms');
subplot(3,1,3); plot(tm,lsb); grid on; title('LSB(t)'); xlabel('ms');

% 5) Magnitude spectra over [-20k, 20k]
figure('Name','Spectra');
subplot(3,1,1); plot(f,abs(spec(m)));   xlim([-20000 20000]); grid on; ti-
tle('|M(f)|');
subplot(3,1,2); plot(f,abs(spec(usb))); xlim([-20000 20000]); grid on; ti-
tle('|SSB\_USB(f)|');
subplot(3,1,3); plot(f,abs(spec(lsb))); xlim([-20000 20000]); grid on; ti-
tle('|SSB\_LSB(f)|');
xlabel('Hz');

% 6) Bandwidth: occupied positive-frequency lines
pk = @(x) f(f>0 & abs(spec(x)) > 0.01*max(abs(spec(x))));
disp('DSB-SC lines (Hz):'); disp(pk(dsb));
disp('SSB-USB lines (Hz):'); disp(pk(usb));
fprintf('BW_DSB = %d Hz, BW_SSB = %d Hz\n', 2*Fm, Fm);

% 7) SSB via Hilbert transform
mh    = imag(hilbert(m));
usb_h = (m.*cos(2*pi*Fc*t) - mh.*sin(2*pi*Fc*t));


% 8) Compare
fprintf('Max |USB_hilbert - USB_filtered| = %.2e\n', max(abs(usb_h-usb)));
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figure('Name','Hilbert vs filtering');
subplot(2,1,1); plot(tm,usb,'b',tm,usb_h,'r--'); grid on; legend('Filtered','Hil-
bert');
title('USB: filtering vs Hilbert'); xlabel('ms');
subplot(2,1,2); plot(f,abs(spec(usb)),'b',f,abs(spec(usb_h)),'r--');
xlim([-20000 20000]); grid on; title('Spectra'); xlabel('Hz');

% 9) Coherent demodulation
d = usb.*c;

% 10) Spectrum before filtering
figure('Name','Demod spectrum');
plot(f,abs(spec(d))); xlim([-25000 25000]); grid on;
title('Spectrum of USB(t)\cdot c(t) before LPF'); xlabel('Hz');

% 12) Manual LPF (ideal, frequency-domain), Fc_cut = 1000 Hz
Fcut = 1000;
D = fftshift(fft(d));
r = real(ifft(ifftshift(D.*(abs(f) <= Fcut+1))));

figure('Name','Recovered');
plot(tm,m,'b',tm,r,'r'); grid on; legend('m(t)','r(t) (unscaled)');
title('Recovered vs original'); xlabel('ms');

% 13) Amplitude comparison and scaling
scale = max(m)/max(r);
fprintf('Amplitude of r(t) = %.3f  -> scale factor = %.2f\n', max(r), scale);
figure('Name','Scaled recovered');
plot(tm,m,'b',tm,scale*r,'r--'); grid on; legend('m(t)','scaled r(t)');
title('After amplitude restoration'); xlabel('ms');

%% ================ PART 2: SSB-TC =================
A    = 2;
sstc = A*c + usb;

% envelope (analytic-signal magnitude)
env = abs(hilbert(sstc));

% 2) time domain with +/- envelopes
figure('Name','SSB-TC');
plot(tm,sstc,'b'); hold on; plot(tm,env,'r',tm,-env,'r'); grid on;
title('SSB\_TC(t) with envelopes'); xlabel('ms');

% 3-4) envelope detection, DC removal and scaling (USB has amplitude m/2)
r2 = 2*(env - mean(env));

% 6) r(t) vs m(t) and its spectrum
figure('Name','SSB-TC recovery');
subplot(2,1,1); plot(tm,m,'b',tm,r2,'r--'); grid on; legend('m(t)','r(t)');
title('Recovered message (envelope detector)'); xlabel('ms');
subplot(2,1,2); plot(f,abs(spec(r2))); xlim([-5000 5000]); grid on;
title('|R(f)|'); xlabel('Hz');
```